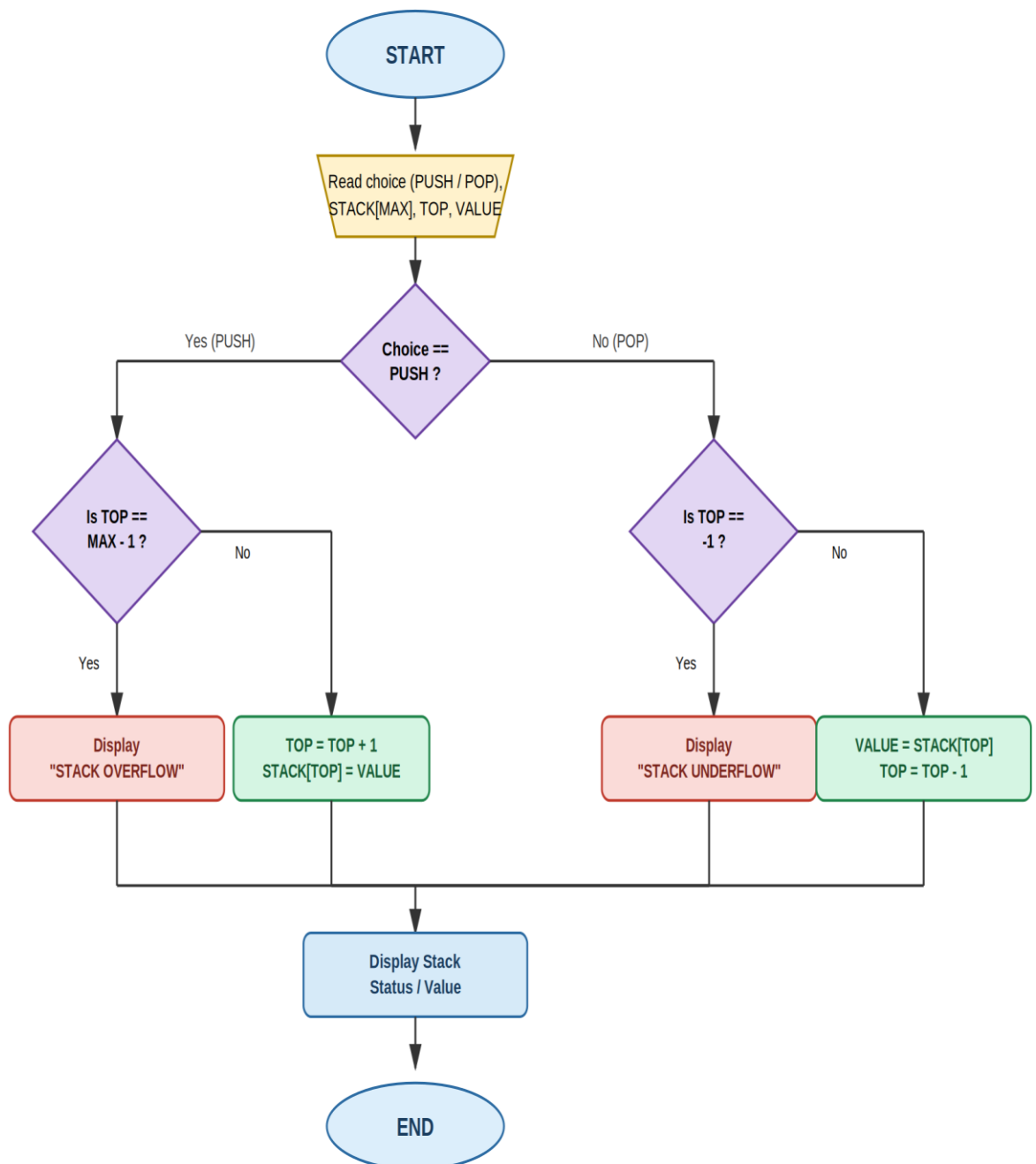


ASSESSMENT TOOL 3 — Flow Chart to Code Conversion

15. Convert a flowchart for checking Stack Overflow and Underflow conditions into a C program

1. Flowchart



2. C Program

```
/* =====
Program : Stack Overflow and Underflow Checking (PUSH / POP)
Description : Implements a stack using an array and checks
               for overflow (on push) and underflow (on pop)
               conditions, as per the corresponding flowchart.
===== */

#include <stdio.h>

#define MAX 5    /* Maximum size of the stack */

int stack[MAX];
int top = -1;    /* top = -1 means stack is empty */

/* Function to push an element onto the stack */
void push(int value)
{
    if (top == MAX - 1)
    {
        printf("STACK OVERFLOW! Cannot push %d. Stack is full.\n", value);
    }
    else
    {
        top = top + 1;
        stack[top] = value;
        printf("Pushed %d onto the stack.\n", value);
    }
}

/* Function to pop an element from the stack */
void pop(void)
{
    int value;

    if (top == -1)
    {
        printf("STACK UNDERFLOW! Stack is empty. Nothing to pop.\n");
    }
    else
    {
        value = stack[top];
        top = top - 1;
        printf("Popped %d from the stack.\n", value);
    }
}

/* Function to display current contents of the stack */
void display(void)
{
    int i;

    if (top == -1)
    {
        printf("Stack is empty.\n");
    }
}
```

```

    else
    {
        printf("Stack elements: ");
        for (i = top; i >= 0; i--)
        {
            printf("%d ", stack[i]);
        }
        printf("\n");
    }
}

int main(void)
{
    int choice, value;

    do
    {
        printf("\n----- STACK MENU ----- \n");
        printf("1. PUSH\n");
        printf("2. POP\n");
        printf("3. DISPLAY\n");
        printf("4. EXIT\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("Enter value to push: ");
                scanf("%d", &value);
                push(value);
                break;

            case 2:
                pop();
                break;

            case 3:
                display();
                break;

            case 4:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 4);

    return 0;
}

```

3. Sample Output (Overflow / Underflow test)

```
Enter your choice: 1
Enter value to push: 60
STACK OVERFLOW! Cannot push 60. Stack is full.
```

```
Enter your choice: 2
(after stack is emptied)
STACK UNDERFLOW! Stack is empty. Nothing to pop.
```